

POSEY FIELD, AL, USA

WMO: 720413

Lat: 34.267N

Lon: 87.600W

Elev: 284

StdP: 97.96

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00138

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.4	(d) -15.9	(e) 1.0	(f) -5.4	(g) -12.8	(h) 1.3	(i) -3.8	(j) 8.6	(k) 8.5	(l) 8.0	(m) 9.1	(n) 2.3	(o) 0	(p) 0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	34.0	24.1	33.0	24.3	32.3	24.2	26.1	31.5	25.6	31.0	25.1	30.3	2.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.4	28.4	24.1	19.6	27.8	23.6	19.1	27.3	82.2	31.5	79.7	31.4	77.6	30.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.8	5.7	-12.5	35.9	3.1	1.7	-14.7	37.1	-16.5	38.1	-18.2	39.0	-20.4	40.2
			-13.1	27.3	3.0	0.5	-15.3	27.6	-17.0	27.9	-18.7	28.2	-20.9	28.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.3	7.0	11.0	16.0	20.1	24.8	25.9	25.3	22.6	16.2	9.5	6.8
	DBStd	8.82	6.26	6.21	5.46	4.31	3.82	2.28	2.19	2.13	3.33	4.75	4.91	5.62
	HDD10.0	589	197	122	56	6	1	0	0	0	0	7	67	134
	HDD18.3	1839	436	319	231	95	28	1	0	0	7	98	267	358
	CDD10.0	2714	20	38	87	184	315	443	492	473	378	199	51	35
	CDD18.3	923	0	2	4	23	84	193	233	214	134	32	2	1
	CDH23.3	8292	0	3	36	216	718	1854	2154	1817	1182	300	12	1
	CDH26.7	3399	0	0	4	28	213	827	982	779	485	79	2	0
Wind	WSAvg	2.2	2.7	3.0	2.9	2.8	2.2	1.7	1.5	1.4	1.4	1.9	2.3	2.7
Precipitation	PrecAvg	1548	147	151	141	133	133	132	132	101	99	87	130	156
	PrecMax	2020	267	266	254	341	370	343	231	221	167	176	242	430
	PrecMin	930	56	66	21	38	19	35	46	4	16	0	78	64
	PrecStd	264	58	64	55	70	96	70	57	54	41	52	43	87
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	23.4	27.2	28.8	32.3	36.1	36.1	36.0	34.5	31.6	24.8	21.8
		MCWB	16.3	18.2	17.8	19.0	22.8	22.3	24.4	25.8	22.2	22.4	16.4	18.5
	2%	DB	17.9	21.1	23.9	27.5	30.5	33.6	33.9	33.6	32.9	28.7	22.2	19.9
		MCWB	15.8	17.0	16.6	18.5	22.2	24.2	24.8	24.6	22.8	20.6	15.2	17.9
	5%	DB	16.2	19.0	22.2	26.1	28.9	32.6	32.8	32.5	31.6	26.9	20.0	17.9
		MCWB	13.9	15.8	15.8	18.0	21.2	24.2	24.9	24.5	23.1	19.2	15.9	16.0
	10%	DB	14.0	17.0	20.0	24.0	27.6	31.3	32.0	31.1	29.7	24.5	17.9	16.2
		MCWB	11.7	13.2	14.6	17.1	20.8	23.7	24.6	24.1	22.7	18.6	13.9	14.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	20.0	19.5	21.5	24.0	26.3	26.8	26.9	25.4	23.9	20.3	19.8
		MCDB	18.8	21.7	23.7	26.0	29.2	31.7	32.0	32.4	29.8	28.1	21.2	20.8
	2%	WB	16.5	18.4	18.2	20.3	23.1	25.6	26.1	26.1	24.6	22.9	18.6	18.0
		MCDB	17.2	20.3	21.8	24.6	28.5	31.4	31.5	31.5	29.5	26.7	20.2	19.1
	5%	WB	14.8	16.8	17.1	19.5	22.4	25.1	25.6	25.5	24.0	21.4	16.8	16.6
		MCDB	16.0	18.4	20.3	23.6	27.5	30.9	31.0	30.6	28.9	24.5	18.1	17.8
	10%	WB	12.6	14.5	16.0	18.6	21.7	24.4	25.2	24.9	23.3	19.9	14.5	14.6
		MCDB	13.5	16.2	19.2	22.1	26.3	29.5	30.4	29.4	27.9	23.2	17.8	16.0
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.0	12.2	12.9	11.7	12.0	11.1	10.8	12.3	13.1	12.7	9.9
		MCDBR	11.5	12.5	14.3	14.3	13.4	13.6	12.6	12.9	14.4	15.1	14.7	10.4
		MCWBR	8.9	8.1	8.2	6.5	5.8	5.3	4.6	5.0	5.6	7.2	9.0	8.2
	5% WB	MCDBR	9.3	9.7	11.5	11.1	11.2	11.7	11.1	10.6	11.5	11.2	10.3	9.5
		MCWBR	8.5	7.5	7.6	6.0	5.4	5.3	4.6	4.7	5.1	6.5	8.5	8.5
Clear-Sky Solar Irradiance	taub	0.312	0.317	0.354	0.386	0.426	0.454	0.495	0.503	0.416	0.362	0.331	0.312	
	taud	2.520	2.503	2.422	2.320	2.252	2.196	2.110	2.081	2.315	2.451	2.480	2.536	
	Ebn at Noon	901	934	920	900	864	835	799	785	844	870	868	880	
	Edh at Noon	83	93	110	127	138	146	158	160	121	97	85	77	
All-Sky Solar Radiation	RadAvg	2.52	3.03	4.18	5.38	5.92	6.26	6.00	5.54	4.84	3.90	2.86	2.15	
	RadStd	0.20	0.39	0.39	0.38	0.45	0.49	0.35	0.33	0.42	0.51	0.30	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (31 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+94

Nomenclature: See separate page